

ITIS 4250
Efren Antonio
Lab #5
November 26, 2024

Overview

On November 26, 2024, Detective Jessica Fletcher of the North Carolina Police Department provided me with a forensic image in EnCase E01 format for analysis. The image was created from a desktop computer belonging to Eugene "Gene" Poole, a known felon suspected of involvement in the manufacture and sale of drugs. Detective Fletcher obtained the computer under a North Carolina District Court Search Warrant as part of an active investigation into Poole's activities and his possible associates. The primary objective of this analysis is to uncover evidence that may link Poole and his associates to drug-related activities, including communications, file transfers, media, and other digital artifacts.

Exam Preparation

The examination environment used is a HP Laptop running Microsoft Windows 11 Home Version 10.0.22631. The tool used for verification of the image is Forensic Tool Kit (FTK) Imager Version 4.7.1.2. The tool used for analyzing communications and web searches is Autopsy Version 4.21.0. The tool to find the OS version was Registry Explorer Version 1.1.0.6

The verified hash of Gene2021.E01 is **75b28a2dd3d0c58fd1b944d1707cba1a** provided in Figure 1 below:

Name	Gene2021.E01
Sector count	5683200
MD5 Hash	
Computed hash	75b28a2dd3d0c58fd1b944d1707cba1a
Stored verification hash	75b28a2dd3d0c58fd1b944d1707cba1a
Verify result	Match

Figure 1: Gene2021.E01 FTK Imager MD5 Hash Verification

Analysis

1. Gene denies the computer is his. Who was the Owner and the Organization to whom it was registered?

The Owner listed in the operating system information in Autopsy is **Eugene Poole** and the Organization is **The Guild of Calamitous Intent**. These can be seen in Figure 2 below:

Temporary Files Directory	%SystemRoot%\TEMP
Path	C:\Windows
Product ID	00329-20000-00001-AA244
Owner	Eugene Poole
Organization	The Guild of Calamitous Intent

Figure 2: Operating System Owner and Organization

2. We believe the location of his mountain hideout might been in a text file. Can you tell us anything about that?

There is a .txt file in the recycle bin named **TheMountains.txt**. This file may have contained information about the location of the mountain hideout but I was not able to access the contents of the file. This can be seen in Figure 3 below:

Recycle Bin							
Table Thumbnail Summary							
Source Name	S	C	O	Path	Time Deleted	Username	Data Source
 SRC97QP5.txt				C:\Users\IEUser\Desktop\TheMountains.txt	2018-09-03 09:36:49 EDT		LogicalFileSet1

Figure 3: TheMountains.txt in Recycle Bin

3. We found a flash drive we believe may have been used with Gene's computer, but it was at the home of his associate. Can you determine what thumb drives may have been attached to Gene's computer to help us get new legal authority?

There was a **Transcend Information Inc. JetFlash** thumb drive that was attached to Gene's computer. The information for this is provided in Figure 4 below:

USB Device Attached								
Table Thumbnail Summary								
Source Name	S	C	O	Date/Time	Device Make	Device Model	Device ID	Data Source
 SYSTEM			0	2020-01-26 13:27:59 EST		ROOT_HUB	48&33a7b1e7&0	LogicalFileSet1
 SYSTEM			0	2020-01-26 13:27:59 EST		ROOT_HUB20	48&25379f10&0	LogicalFileSet1
 SYSTEM			0	2018-09-03 10:28:49 EDT	Transcend Information, Inc.	JetFlash	FWOAJEB	LogicalFileSet1
Type					Value			
Date/Time					2018-09-03 10:28:49 EDT			
Device Make					Transcend Information, Inc.			
Device Model					JetFlash			
Device ID					FWOAJEB			
Source File Path					/LogicalFileSet1//F/Windows/System32/config/SYSTEM			
Artifact ID					-9223372036854775602			

Figure 4: USB Device Attached to Gene's Computer

4. Was he using an office program to make documents? We think some files we got from our informant may have been created by Gene. Any idea how often he used the software?

5. Gene may have had an interest in Cybersecurity and "hacking", even though his strengths are in chemistry, bio-engineering and machining. Is there any evidence he may have tried to use or learn anything about hacking tools?

6. Gene may have used a phone to photograph a secure location, according to our source. Any idea what? What happened to the device?

7. Any other devices he might have used to photograph sensitive technology?

8. Gene has mentioned in recorded conversations he has a stash of drugs he keeps at a beach house just in case his lab gets raided. Is there anything that might tell us he is familiar with particular beaches, or towns near beaches?

Conclusion

The forensic analysis of Malcom Monarch's laptop and the Clarion Hotel Business Center computer provided key insights into his activities. It revealed communications with a Guild operative along with searches related to fireworks, bombs, and using tools like CCleaner to remove traces of activity. Monarch also explored forensic software, potentially to understand how evidence might be uncovered. Downloaded images and web history offer details about his appearance and browsing behavior. These findings provide valuable information for further examination of Monarch's actions and intent.

Name	Gene2021.E01
Sector count	5683200
MD5 Hash	
Computed hash	75b28a2dd3d0c58fd1b944d1707cba1a
Stored verification hash	75b28a2dd3d0c58fd1b944d1707cba1a
Verify result	Match

Figure 17: Reverified MD5 Hash Values After The Examination

Signed:

Efren Antonio